Nazreen Shah Ambalath

Ph.D. Candidate
Indraprastha Institute of Information Technology (IIIT Delhi), India
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I am a Ph.D. candidate at IIIT Delhi and a ***Prime Minister's Fellow for Doctoral Research*** (SERB-FICCI), working at the intersection of federated learning, distributed optimization, and representation learning. My research focuses on building theoretically grounded and communication-efficient federated learning algorithms, with contributions spanning personalized federated learning, continual federated learning, decentralized learning, and over-the-air aggregation. I am particularly interested in understanding convergence behavior under practical constraints such as heterogeneity, limited communication bandwidth, and streaming data. My work has been published at venues including ICASSP, AISTATS, CoLLAs, and TMLR. I have also presented a poster at ICDCN 2025, which won the Best Poster Award. I was also recently a Visiting Scholar at Wayne State University, collaborating on continual learning and unlearning techniques in large-scale Neural Networks.

Education

Ph.D., Department of ECE, Indraprastha Institute of Information Technology (IIIT Delhi), India 2022 – Present
Expected graduation: Jan 2027 CGPA: 8.85*
Thesis supervisor: Dr. Ranjitha Prasad
Research Areas: Federated Learning, Personalized Federated Learning, Continual Federated Learning, Unsupervised Federated Learning, Decentralized Federated Learning

Integrated M.Tech in Computer Science, University of Hyderabad, Telangana, India 2015 – 2020
First Class

Research Interests

- Federated Learning (Personalized, Continual, Unsupervised, Decentralized)
- Over-the-Air Federated Learning and Robust Aggregation
- Continual Learning-theoretical foundations and distributed settings
- Machine Unlearning in Large Vision-Language Models

Research Experience

Visiting Scholar, Wayne State University, Detroit, USA Feb 2026 - Jul 2026
Supervisor: Dr. Prashant Khanduri, Department of Computer Science

Research Assistant, Indraprastha Institute of Information Technology, Delhi, India Sep 2021 - Jan 2022
Supervisor: Dr. Ranjitha Prasad, ECE Department
Project funded by DST-SERB, India: "Personalized Federated Learning"

Research Associate, Indian Statistical Institute, Kolkata, India Sep 2020 - Jun 2021
Supervisor: Prof. Ashish Ghosh, Machine Intelligence Unit
Project funded by DRDO, India: "Hardware Trojan Detection on PCB using X-ray Images"

M.Tech Thesis, Institute for Development and Research in Banking Technologies, Hyderabad, India Jun 2019 - Aug 2020
Supervisor: Dr. Rajarshi Pal, Centre for Excellence in Cyber Security
Project: "Reversible Data Hiding in Color Images"

Teaching Experience

Teaching Assistant - Distributed and Federated Machine Learning, IIIT Delhi 2023, 2025

Teaching Assistant - Probability and Random Process, IIIT Delhi 2024

Teaching Assistant - Convex Optimization, IIIT Delhi 2023

Peer-Reviewed Publications

Shah N, Somireddy N, Shaban Z, Prasad R, Bharath B.N. (2026). Federated Continual Learning as a Distributed Drift-Plus-Penalty Control Problem. Conference on Lifelong Learning Agents, CoLLAs 2026 [Oral Presentation (Top 8%)].

Shah N, Arya G, Bharath B.N., Prasad R. (2026). Theoretical Foundations of Continual Learning via Drift-Plus-Penalty. *Transactions on Machine Learning Research (TMLR 2026).*

Chugh S, Prasad R, **Shah N**. (2026). Informative Perturbation Selection for Uncertainty-Aware Post-hoc Explanations. *ECML-PKDD 2026*.

Fathima Z, Shaban Z, Jamala Y, **Shah N**, Bharath B.N., Prasad R. (2026). Channel-Adaptive Robust Aggregation for Over-the-Air Federated Learning in Heterogeneous Networks. *ICASSP 2026 - IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE.

Arora K*, **Shah N***, Sethi A, Prasad R, Uliyar M, Krishna A.G. (2025). Personalized Federated Autoencoders: Unsupervised Representation Learning at the Edge. *16th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2025)*.

Keshri S.K., **Shah N**, Prasad R. (2025). On the Convergence of Continual Federated Learning Using Incrementally Aggregated Gradients. *28th International Conference on Artificial Intelligence and Statistics (AISTATS)*.

Shaban Z, **Shah N**, Prasad R. (2025). Robust Over-The-Air Federated Learning In Heterogeneous Networks. *ICASSP 2025 - IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1–5. IEEE.

Shah N, Prasad R. (2024). Federated Probabilistic Representation Learning using Information Geometric Approach. *Proceedings of the 8th International Conference on Data Science and Management of Data (12th ACM IKDD CODS and 30th COMAD)*, pp. 341–343.

Keshri S.K., **Shah N**, Prasad R. (2024). Continual Federated Learning with Aggregated Gradients. *Proceedings of the 8th International Conference on Data Science and Management of Data (12th ACM IKDD CODS and 30th COMAD)*, pp. 347–349.

Shah N, Goyal P, Prasad R. (2024). Importance Sampling Based Federated Unsupervised Representation Learning. *ICASSP 2024 - IEEE International Conference on Acoustics, Speech and Signal Processing*, Seoul, Korea, pp. 7565–7569.

Madhan-Sohini A, Dominic D, **Shah N**, Prasad R. (2023). Over-the-air clustered wireless federated learning. *IEEE Globecom Workshops (GC Wkshps)*, pp. 203–208. IEEE.

Preprints and Manuscripts Under Review

Shaban Z, **Shah N**, Prasad R. (2026). Unified Robust and Convergent Over-The-Air Federated Learning under Channel Noise and Heterogeneity. *[Under Review]*

Arya G, **Shah N**, Bharath B.N., Prasad R. (2026). Stochastic Gradient Drift-Plus-Penalty for Replay-based Continual Learning. *[Under Review]*

Sarker A, **Shah N**, Sultan R, Jang R, Zhu D, Khanduri P. (2026). Partition-Aware Unlearning for Removing Spurious Correlations in Large Vision-Language Models. *[Under Review]*

Conference Presentations & Professional Activities

Selected to attend and present at **MLSS** Machine Learning Summer School 2026, Columbia University (acceptance rate: 16.67%), NYC, USA. Jun 15–26, 2026.

Selected to attend and present at **CVPR** Women in CV (WiCV) Workshop 2026, Denver, USA. Jun 3–7, 2026.

Selected to attend and present at **IPDPS** PhD Forum 2026, New Orleans, USA. May 25–29, 2026.

Selected to attend and present a poster at **IndoML** Summer School 2025, BITS Pilani. Dec 19–21, 2025.

Selected to attend **M2L** Summer School 2025 (acceptance rate $<$18%), Split, Croatia. Sep 8–12, 2025.

Microsoft Research India Academic Summit 2025, Virtual.

Work accepted at sister conference track, **CoLLAs 2025**, Philadelphia, USA.

Paper presented at **AISTATS 2025**, Phuket, Thailand.

Paper presented at **ICASSP 2025**, Hyderabad.

Poster presented at **ICASSP 2025** Pre-conference Event, IIT Delhi.

Poster accepted at **ICDCN 2025**, IIT Hyderabad.

Paper accepted at **CODS-COMAD Dec '24**, IIT Jodhpur - YRS Track.

Abstract selected as a poster at **WiML Workshop - NeurIPS 2024**, Vancouver, Canada.

Selected to attend **EEML** Summer School 2024 (acceptance rate $<$30%), Novi Sad, Serbia, Jul 15–21, 2024. Work presented as poster.

Selected to attend **OxML** Summer School virtually, Jul 1–5, 2024.

Attended **IEEE ICASSP 2024**, Seoul, South Korea. Presented "Importance Sampling Based Federated Unsupervised Representation Learning." Apr 14–19, 2024.

Attended **CODS-COMAD**, January 2024, IIIT Bangalore.

Attended **IEEE GlobeCom Workshops**, Kuala Lumpur, Malaysia. Presented "Over-the-air clustered wireless federated learning." Dec 3–8, 2023.

Presented poster on "Clustered wireless federated learning" at **RIISE** 2023 Research Expo.

Presented at **IEEE SPS Seasonal School** on AI for Optimisation in Signal Processing Research Symposium on "Decentralized Federated Learning."

Took part in Annual Cyber Drill for Banks at **IDRBT** (2020).

Co-organized a **Hackweek** in SCIS, University of Hyderabad (2019).

Participated in workshops for System Lab Administration and Campus Network Facilities (2017).

Service

Reviewer - IEEE Transactions on Mobile Computing (TMC)	2026
Reviewer - IEEE Signal Processing Letters (SPL)	2026
PC Member - MLSys (Artifact Evaluation)	2026
Reviewer - Doctoral Symposium, MMSys	2026
Reviewer - IEEE ICC	2026
Reviewer - IEEE ICASSP	2026
Program Committee Member - AAAI FLCA Workshop	2025
Reviewer - WiML-NeurIPS	2024, 2025
Volunteer Organizer - NCC 2025, IIT Delhi	2025

Community Service

Joint Secretary, Student Organization	2020
Hostel Chairperson, University of Hyderabad	2019
Mess Secretary, University of Hyderabad	2018
Department Website Developer, Department of Sociology, University of Hyderabad	2018
Co-ordinator, University Fest (Sukoon) and Community Fest (Keraleeyam), University of Hyderabad	2017
Co-organizer, Hackweek, SCIS, University of Hyderabad	2019

Fellowships, Awards, and Honors

DAAD AInet Fellowship (Postdoc-NeT-AI, Reliable AI cohort), German Academic Exchange Service	2026
Recognized as top 100 AI/ML researchers in India by Amazon	2026
Dean's List of Best Teaching Assistant, IIIT Delhi	2026
Financial aid to attend ECML-PKDD, 2026, Naples, Italy	2026
Financial aid to attend CoLLAs, 2026, Bucharest, Romania	2026
Financial aid to attend MLSS, Columbia University, NYC	2026
WiCV Travel Grant to attend CVPR 2026, Denver, USA	2026
IEEE TCPP Travel Grant to attend IPDPS 2026, New Orleans, USA	2026
Overseas Research Fellowship, IIIT Delhi (International Lab Visit)	2026
Dean's List of Best Teaching Assistant, IIIT Delhi	2025
Full grant to attend M2L Summer School 2025, Croatia	2025
Financial aid to attend AISTATS 2025, Thailand	2025
Best Poster Award at ICDCN 2025	2025
Qualcomm Travel Grant to attend ICDCN 2025, IIT Hyderabad	2024
Travel Grant to attend CODS COMAD Dec 2024, IIT Jodhpur	2024
WiML Corporate Travel Grant to attend NeurIPS 2024, Vancouver, Canada	2024
Full grant to attend EEML 2024, Serbia	2024
IEEE SPS Travel Grant to attend IEEE ICASSP 2024, Seoul, South Korea	2024
Travel Grant to attend CODS-COMAD 2024, IIIT Bangalore	2024
Prime Minister's Fellowship for Doctoral Research, SERB-FICCI (Industry partner - LightMetrics)	2023
Junior Research Fellowship, DST-SERB	2021
UGC-NET Qualified	2019
GATE Qualified	2019
Mathematics School Topper (Gold Medal)	2014
Computer-Maths Stream Topper (Gold Medal)	2014

Professional Affiliations

IEEE, Member	2023 - Present
IEEE Signal Processing Society (SPS), Member	2024 - Present
IEEE Women in Engineering (WIE), Member	2024 - Present
IEEE Young Professionals, Member	2024 - Present

Skills

- **Programming:** Python, PyTorch, SciPy, MATLAB, LaTeX, C/C++
- **Technical:** Convex Optimization, Bayesian Machine Learning, Classification, Deep Learning, Linear Algebra, Federated Learning
- **Tools:** Git, Linux, Weights & Biases, Chameleon, Docker
- **Soft Skills:** Team Work, Agile, Observant, Responsible, Hard-Working

Languages

- English - Proficient
- Malayalam - Native
- Tamil - Intermediate
- Hindi - Intermediate
- Arabic - Basic

Referees

Dr. Ranjitha Prasad
Assistant Professor, ECE
IIIT Delhi, India
Federated Learning,
Continual Learning
ranjitha@iiitd.ac.in

Dr. Prashant Khanduri
Assistant Professor, CSE
Wayne State University, USA
Large-Scale Models,
Federated Learning
prashant.khanduri@wayne.edu

Dr. Bharath B.N.
Assistant Professor, EE
IIT Dharwad, India
Federated Learning,
Optimization
bharathbn@iitdh.ac.in

Additional references available upon request.

Last updated: October 24, 2026